

Vishnu Vardhan Mandula

United States

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SUMMARY

Software Engineer with 4+ years of experience delivering scalable backend systems, APIs, and data platforms across healthcare, finance, and manufacturing domains. Designed and deployed AWS and Databricks powered solutions that processed 2.3M+ records/day, reduced fraud detection time by 65% and cut deployment cycles by 40%. Skilled in Python, Core Java, SQL, Kafka, Redis, Rocks DB, Hazel and cloud-native architectures, with a track record of optimizing pipelines and integrating Generative AI into production workflows.

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, JavaScript, TypeScript, Golang (Go), C++
- **Web Stack:** HTML, CSS, React, Node.js, Spring Boot, Spring MVC, Angular, Django, Fast API, REST API's, GraphQL, JSON, jQuery
- **Cloud & DevOps:** AWS (Lambda, EC2, S3, IAM, CloudWatch), Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, CI/CD
- **Databases & Messaging:** PostgreSQL, MongoDB, DynamoDB, Redis, Kafka, Rocks DB, Hazel cast, RabbitMQ, Big Query, MySQL, NoSQL
- **Security:** OAuth2, JWT, Snyk, Trivy, IAM policies, CI/CD vulnerability scanning, RBAC
- **Tools:** Elasticsearch, Kibana, DataDog, Grafana, Git, Jira, Maven, Google Protocol Buffers
- **Software Engineering Practices:** Agile/Scrum, TDD, Object-Oriented Design, Design Patterns, Performance Optimization

EXPERIENCE

Freddie Mac

Software Engineer

Jan 2024 – Present

Richmond, VA

- Architected Core Java(Spring Boot) microservices with Kafka + Hazelcast for distributed telemetry, processing 2.3M records/day at 99% accuracy.
- Optimized low latency data retrieval by integrating Rocks DB as an embedded key-value store, cutting query response time by 35%.
- Constructed RESTful APIs with IAM-based access controls, facilitating secure data transactions for 5+ external fintech partners.
- Streamlined deployment processes using GitHub Actions and Docker, reducing release time by 40% while maintaining a 100% success rate.
- Instrumented monitoring with CloudWatch and Grafana, boosting system uptime from 98.7% to 99.99% via real-time alerting.

Freddie Mac

Software Developer Intern

Jun 2023 – Dec 2023

Richmond, VA

- Refactored Spring Boot services into containers, decreasing API latency by 40% and memory consumption by 30% through JVM tuning.
- Devised CI/CD automation with GitHub Actions and Docker, increasing deployment velocity by 50% across test environments.
- Integrated Okta for RBAC and OAuth2 protocols, preventing unauthorized access across secured API endpoints.
- Executed unit and integration tests using JUnit and Selenium, raising test coverage to 95% and lowering production bugs by 38%.
- Documented system architecture and conducted code reviews, accelerating onboarding for new team members by 25%.

Thryve Digital Health

Software Developer

Jun 2020 – Jul 2022

Hyderabad, IND

- Devised ETL pipelines with Google Cloud Functions and Big Query, processing 10+ GB of advertising data daily with 99.9% consistency.
- Programmed Core Java utilities for financial data aggregation, leveraging Google Protocol Buffers for data exchange across 4 business units.
- Assembled automation test suites with Selenium and Python for UI regression, decreasing deployment errors by 30%.
- Deployed observability stack using Kibana and Datalog, minimizing incident response from 15 to under 5 minutes.

Cliff.AI

Software Developer

Dec 2019 – Apr 2020

Hyderabad, IND

- Engineered backend crawlers and APIs in Python and Java to process structured/unstructured business data.
- Automated ingestion from APIs like Google Analytics and Salesforce; improved update frequency by 50%.
- Created internal microservices using Flask and launched them with Docker and Jenkins CI.

EDUCATION

M.S. in Computer Science Stevens Institute of Technology, Hoboken, NJ. GPA: 3.8/4.0

Aug 2022 – Dec 2023

B. Tech in Computer Science CVR College of Engineering, Hyderabad, India. GPA: 3.8/4.0

PROJECTS

DevSecOps AI Application Pipeline

Mar 2025 - May 2025

- Assembled full-stack application with React frontend and Java backend, deploying to AWS with Terraform; supported over 1,000 concurrent users.
- Embedded Trivy, Snyk, and Aqua security scanners in CI pipeline, identifying 50+ critical vulnerabilities pre-deployment.
- Enabled automated secret detection, dependency analysis, and Slack alerts, cutting incident response time by 60%.

Cloud-Native Course Management System

Sep 2024 - Nov 2024

- Developed stateless REST APIs in Node.js/Express, backed by Sequelize and PostgreSQL, implemented with Auto Scaling and ELB.
- Designed secure CI/CD pipelines with GitHub Actions and IAM roles, cutting deployment failures by 70% across environments.
- Established CloudWatch logging and custom alert rules to uphold 99.99% system uptime.

Travel Booking Web Application

Jan 2024 - Mar 2024

- Constructed Spring Boot services with Kafka and Google protocol buffers for serialization to process 10,000+ real-time inventory updates daily.
- Leveraged AWS Lambda for scalable fare computations, supporting 1 million+ pricing queries with millisecond latency.
- Connected Stripe API for secure transactions, reducing monthly chargebacks by 25% via fraud detection logic.

Distributed Authentication Gateway System

Sep 2023 - Dec 2023

- Built Go authentication gateway processing 10,000+ req/sec with less than 10ms latency using JWT/RBAC for 100+ microservices.
- Designed stateless architecture with Hazel cast caching, ensuring 99.9% uptime and 20% faster auth lookups for 1M+ daily authentications.

